

SUPPLEMENTAL MATERIAL

MONITORING VS. MODELED EXPOSURE DATA IN TIME-SERIES STUDIES OF AMBIENT AIR POLLUTION AND ACUTE HEALTH OUTCOMES

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Supplemental Table S1. US Environmental Protection Agency and SEARCH monitoring coverage within the 20-county Atlanta study region.

Site ID	Lat	Long	CO	NO ₂	SO ₂	O ₃	PM _{2.5}	SO ₄	EC	OC
JST ^{1,2}	33.77753	-84.4167	X	X	X	X	X	X	X	X
YRK ¹	33.92833	-85.0456	X	X	X	X	X	X	X	X
130630091	33.61036	-84.3909					X			
130670003	34.01546	-84.6074				X	X			
130670004	33.89932	-84.6615					X			
130770002	33.40405	-84.7459				X	X			
130890002	33.68796	-84.2904	X	X	X	X	X	X	X	X
130892001	33.90139	-84.2799					X			
130970004	33.74245	-84.7778				X				
131210032	33.81956	-84.3897					X			
131210039	33.80229	-84.4356					X			
131210048	33.77933	-84.3958		X	X					
131210055	33.72043	-84.3572			X	X	X			
131210099	33.87647	-84.3803	X							
131350002	33.96224	-84.0691				X	X			
131390003	34.29969	-83.8128					X			
131510002	33.43376	-84.1618				X	X			
132230003	33.9285	-85.0453								
132470001	33.59044	-84.0664		X		X				

¹ Jefferson St. (JST) and Yorkville (YRK) sites are part of the SouthEastern Aerosol Research and Characterization (SEARCH) network

² JST was selected as the central monitor (CM) metric for this study.

Supplemental Table S2. Within-pollutant Pearson correlations between daily exposure metrics.^{1, 2, 3}

Pollutant	Metric	CM	SA	IDW	KRIG	CMAQ	CMAQKRIG	CMAQDF	LUR	AOD
1-hr max CO	CM	1	0.844	0.873	0.750	0.447	0.600	0.620	-	-
	SA	0.844	1	0.915	0.824	0.502	0.666	0.752	-	-
	IDW	0.998	0.869	1	0.932	0.495	0.760	0.785	-	-
	KRIG	0.993	0.881	0.998	1	0.462	0.835	0.758	-	-
	CMAQ	0.568	0.611	0.586	0.594	1	0.621	0.683	-	-
	CMAQKR	0.998	0.861	0.999	0.997	0.590	1	0.805	-	-
	CMAQDF	0.681	0.850	0.723	0.749	0.650	0.719	1	-	-
1-hr max NO ₂	CM	1	0.797	0.866	0.836	0.460	0.632	0.541	0.588	-
	SA	0.797	1	0.894	0.871	0.486	0.693	0.721	0.663	-
	IDW	0.998	0.811	1	0.968	0.521	0.758	0.728	0.672	-
	KRIG	1.000	0.805	0.998	1	0.533	0.789	0.739	0.679	-
	CMAQ	0.555	0.482	0.562	0.557	1	0.778	0.750	0.864	-
	CMAQKR	1.000	0.802	0.998	1.000	0.564	1	0.896	0.806	-
	CMAQDF	0.642	0.756	0.664	0.646	0.586	0.646	1	0.788	-
1-hr max SO ₂	LUR	0.695	0.685	0.703	0.699	0.887	0.703	0.673	1	-
	CM	1	0.856	0.868	0.818	0.334	0.583	0.472	-	-
	SA	0.856	1	0.892	0.847	0.462	0.619	0.612	-	-
	IDW	0.996	0.872	1	0.957	0.426	0.703	0.633	-	-
	KRIG	1.000	0.866	0.997	1	0.411	0.738	0.608	-	-
	CMAQ	0.468	0.571	0.479	0.476	1	0.547	0.729	-	-
	CMAQKR	1.000	0.860	0.997	1.000	0.476	1	0.754	-	-
8-hr max O ₃	CMAQDF	0.657	0.769	0.693	0.663	0.586	0.663	1	-	-
	CM	1	0.950	0.948	0.927	0.801	0.850	0.900	-	-
	SA	0.950	1	0.974	0.956	0.767	0.887	0.920	-	-
	IDW	0.999	0.958	1	0.992	0.807	0.920	0.950	-	-
	KRIG	0.999	0.958	0.999	1	0.799	0.931	0.948	-	-
	CMAQ	0.787	0.740	0.789	0.786	1	0.732	0.872	-	-
	CMAQKR	1.000	0.953	0.999	0.999	0.792	1	0.901	-	-
24-hr avg PM _{2.5}	CMAQDF	0.917	0.910	0.922	0.919	0.895	0.921	1	-	-
	CM	1	0.948	0.937	0.904	0.633	0.882	0.873	0.723	0.897
	SA	0.948	1	0.978	0.953	0.591	0.933	0.911	0.724	0.949
	IDW	0.974	0.951	1	0.988	0.610	0.964	0.919	0.731	0.945
	KRIG	0.974	0.945	0.999	1	0.592	0.976	0.904	0.715	0.925

Pollutant	Metric	CM	SA	IDW	KRIG	CMAQ	CMAQKRIG	CMAQDF	LUR	AOD
	CMAQ	0.671	0.612	0.665	0.664	1	0.623	0.671	0.865	0.646
	CMAQKR	0.995	0.953	0.991	0.991	0.673	1	0.907	0.733	0.921
	CMAQDF	0.828	0.841	0.857	0.854	0.657	0.846	1	0.769	0.912
	LUR	0.756	0.741	0.754	0.751	0.891	0.760	0.740	1	0.776
	AOD	0.924	0.953	0.925	0.918	0.712	0.929	0.842	0.809	1
24-hr avg SO ₄	CM	1	0.967	0.975	0.969	0.687	0.926	0.676	-	-
	SA	0.967	1	0.981	0.974	0.699	0.930	0.693	-	-
	IDW	1.000	0.968	1	0.995	0.702	0.950	0.696	-	-
	KRIG	1.000	0.973	1.000	1	0.702	0.955	0.695	-	-
	CMAQ	0.695	0.702	0.696	0.699	1	0.750	0.827	-	-
	CMAQKR	1.000	0.969	1.000	1.000	0.699	1	0.735	-	-
	CMAQDF	0.662	0.677	0.664	0.664	0.760	0.664	1	-	-
24-hr avg EC	CM	1	0.858	0.923	0.919	0.598	0.740	0.652	-	-
	SA	0.858	1	0.897	0.892	0.642	0.784	0.720	-	-
	IDW	1.000	0.862	1	0.989	0.631	0.843	0.720	-	-
	KRIG	1.000	0.868	1.000	1	0.631	0.841	0.717	-	-
	CMAQ	0.710	0.693	0.712	0.713	1	0.755	0.755	-	-
	CMAQKR	0.999	0.865	0.999	0.999	0.721	1	0.746	-	-
	CMAQDF	0.675	0.715	0.681	0.678	0.676	0.678	1	-	-
24-hr avg OC	CM	1	0.945	0.942	0.923	0.608	0.845	0.814	-	-
	SA	0.945	1	0.970	0.956	0.632	0.893	0.865	-	-
	IDW	1.000	0.949	1	0.991	0.656	0.919	0.885	-	-
	KRIG	0.999	0.953	1.000	1	0.650	0.930	0.878	-	-
	CMAQ	0.661	0.674	0.665	0.662	1	0.677	0.712	-	-
	CMAQKR	0.999	0.949	0.999	0.999	0.658	1	0.852	-	-
	CMAQDF	0.814	0.864	0.820	0.817	0.678	0.814	1	-	-

¹ Blue shading = correlations among daily exposure metrics for a given pollutant, for the Central ZIP code exposure data

² Red shading = average ZIP code level correlations among daily exposure metrics for a given pollutant, for the Full Spatial information exposure data

³ Exposure metrics included: Central Monitor (CM), Site Average (SA), Inverse Distance Weighting (IDW), Kriging (KRIG), Community Multiscale Air Quality model (CMAQ), CMAQ-kriging adjustment with observations (CMAQKR), CMAQ combined with monitoring data in a data fusion (CMAQDF), Land Use Regression (LUR), and satellite Aerosol Optical Depth (AOD).

Supplemental Table S3. Associations of air pollution and emergency department visits for cardiovascular and respiratory diseases, comparing results among 8 pollutants, 9 exposure metrics, and 3 study approaches.^{1, 2, 3}

Approach:	Cardiovascular ED Visits			Respiratory ED Visits		
	1. CM	2. Central ZIP	3. Full Spatial	1. CM	2. Central ZIP	3. Full Spatial
RR (95% CI) for CO per 250 ppb increase						
CM	1.013 (1.004-1.023)	--	--	1.003 (0.997-1.008)	--	--
SA	--	1.012 (1.000-1.025)	1.012 (1.000-1.025)	--	1.002 (0.995-1.009)	1.002 (0.995-1.009)
IDW	--	1.014 (1.004-1.024)	1.010 (0.998-1.022)	--	1.003 (0.997-1.008)	1.000 (0.993-1.006)
KRIG	--	1.015 (1.004-1.025)	1.009 (0.998-1.020)	--	1.003 (0.997-1.009)	1.001 (0.995-1.006)
CMAQ	--	1.014 (1.003-1.024)	1.002 (0.991-1.014)	--	1.001 (0.995-1.008)	0.998 (0.992-1.004)
CMAQKR	--	1.014 (1.004-1.024)	1.001 (0.990-1.012)	--	1.003 (0.997-1.008)	0.992 (0.986-0.997)
CMAQDF	--	1.011 (1.002-1.021)	1.001 (0.990-1.011)	--	1.005 (0.999-1.010)	1.003 (0.997-1.008)
RR (95% CI) for NO₂ per 10 ppb increase						
CM	1.010 (0.999-1.021)	--	--	0.996 (0.989-1.002)	--	--
SA	--	1.014 (1.000-1.028)	1.014 (1.000-1.028)	--	0.996 (0.988-1.005)	0.996 (0.988-1.005)
IDW	--	1.011 (1.000-1.022)	1.016 (1.002-1.030)	--	0.995 (0.989-1.002)	1.018 (1.010-1.026)
KRIG	--	1.011 (0.999-1.022)	1.020 (1.005-1.034)	--	0.996 (0.989-1.002)	1.026 (1.018-1.034)
CMAQ	--	1.010 (0.999-1.021)	0.996 (0.987-1.005)	--	1.010 (1.003-1.016)	1.012 (1.008-1.017)
CMAQKR	--	1.010 (0.999-1.021)	1.001 (0.990-1.013)	--	0.996 (0.989-1.002)	1.003 (0.997-1.009)
CMAQDF	--	1.013 (1.001-1.025)	0.995 (0.984-1.005)	--	1.012 (1.005-1.019)	1.014 (1.008-1.020)
LUR	--	1.018 (0.996-1.040)	0.992 (0.974-1.010)	--	1.023 (1.010-1.036)	1.025 (1.016-1.035)
RR (95% CI) for SO₂ per 4 ppb increase						
CM	1.003 (0.995-1.011)	--	--	0.997 (0.993-1.002)	--	--
SA	--	1.003 (0.990-1.016)	1.003 (0.990-1.016)	--	1.003 (0.994-1.011)	1.003 (0.994-1.011)
IDW	--	1.003 (0.995-1.012)	1.011 (1.000-1.022)	--	0.997 (0.992-1.002)	1.002 (0.996-1.009)
KRIG	--	1.003 (0.995-1.012)	1.012 (1.002-1.023)	--	0.998 (0.992-1.003)	1.007 (1.001-1.012)
CMAQ	--	1.025 (1.002-1.049)	1.012 (0.992-1.033)	--	1.028 (1.012-1.045)	1.015 (1.004-1.028)

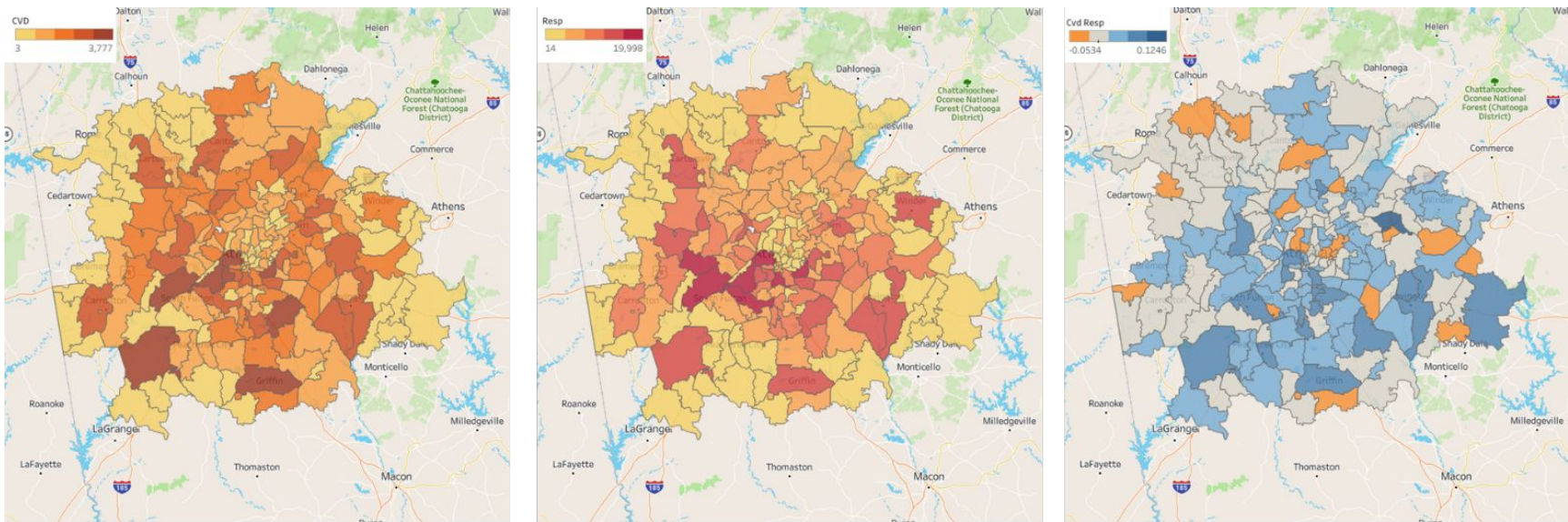
Approach:	Cardiovascular ED Visits			Respiratory ED Visits		
	1. CM	2. Central ZIP	3. Full Spatial	1. CM	2. Central ZIP	3. Full Spatial
CMAQKR	--	1.003 (0.995-1.011)	1.006 (1.000-1.013)	--	0.998 (0.993-1.003)	1.003 (1.000-1.007)
CMAQDF	--	1.007 (0.991-1.023)	0.995 (0.981-1.009)	--	0.999 (0.990-1.009)	1.006 (0.998-1.014)
RR (95% CI) for O ₃ per 15 ppb increase						
CM	1.000 (0.985-1.016)	--	--	1.019 (1.009-1.029)	--	--
SA	--	1.007 (0.990-1.024)	1.007 (0.990-1.024)	--	1.017 (1.006-1.028)	1.017 (1.006-1.028)
IDW	--	1.001 (0.985-1.017)	1.000 (0.984-1.016)	--	1.019 (1.008-1.029)	1.045 (1.035-1.055)
KRIG	--	1.001 (0.985-1.017)	1.001 (0.986-1.016)	--	1.018 (1.008-1.029)	1.038 (1.028-1.047)
CMAQ	--	1.002 (0.986-1.019)	0.997 (0.981-1.013)	--	1.023 (1.011-1.034)	1.030 (1.021-1.040)
CMAQKR	--	1.000 (0.985-1.016)	1.000 (0.987-1.013)	--	1.019 (1.008-1.029)	1.024 (1.017-1.032)
CMAQDF	--	0.999 (0.982-1.016)	0.995 (0.980-1.011)	--	1.015 (1.003-1.026)	1.019 (1.010-1.029)
RR (95% CI) for PM _{2.5} per 5 µg/m ³ increase						
CM	1.017 (1.004-1.030)	--	--	1.001 (0.993-1.009)	--	--
SA	--	1.016 (1.004-1.029)	1.016 (1.004-1.029)	--	1.002 (0.994-1.010)	1.002 (0.994-1.010)
IDW	--	1.018 (1.005-1.031)	1.020 (1.008-1.032)	--	1.000 (0.992-1.008)	1.005 (0.998-1.013)
KRIG	--	1.018 (1.005-1.031)	1.019 (1.007-1.030)	--	1.000 (0.992-1.008)	1.004 (0.998-1.011)
CMAQ	--	1.014 (1.006-1.022)	1.011 (1.002-1.020)	--	1.003 (0.999-1.008)	1.000 (0.995-1.005)
CMAQKR	--	1.018 (1.005-1.030)	1.016 (1.005-1.028)	--	1.000 (0.992-1.008)	1.003 (0.997-1.010)
CMAQDF	--	1.023 (1.009-1.037)	1.022 (1.006-1.037)	--	0.998 (0.989-1.007)	1.005 (0.995-1.014)
LUR	--	1.025 (1.009-1.041)	1.021 (1.004-1.038)	--	1.004 (0.995-1.013)	0.998 (0.989-1.008)
AOD	--	1.014 (1.003-1.025)	1.019 (1.007-1.030)	--	1.005 (0.999-1.011)	1.006 (0.999-1.012)
RR (95% CI) for SO ₄ per 1 µg/m ³ increase						
CM	1.001 (0.992-1.011)	--	--	0.991 (0.985-0.997)	--	--
SA	--	1.000 (0.990-1.010)	1.000 (0.990-1.010)	--	0.994 (0.988-1.000)	0.994 (0.988-1.000)
IDW	--	1.001 (0.992-1.011)	1.001 (0.991-1.011)	--	0.991 (0.985-0.997)	0.995 (0.989-1.001)
KRIG	--	1.001 (0.991-1.011)	1.003 (0.993-1.013)	--	0.991 (0.985-0.997)	0.995 (0.989-1.002)

Approach:	Cardiovascular ED Visits			Respiratory ED Visits		
	1. CM	2. Central ZIP	3. Full Spatial	1. CM	2. Central ZIP	3. Full Spatial
CMAQ	--	1.003 (0.992-1.014)	0.999 (0.988-1.011)	--	0.995 (0.988-1.002)	0.995 (0.988-1.002)
CMAQKR	--	1.001 (0.992-1.011)	1.000 (0.990-1.010)	--	0.991 (0.985-0.997)	0.992 (0.987-0.998)
CMAQDF	--	0.998 (0.989-1.007)	0.995 (0.986-1.004)	--	0.995 (0.990-1.001)	0.991 (0.985-0.996)
RR (95% CI) for EC per 0.5 µg/m³ increase						
CM	1.017 (1.004-1.030)	--	--	0.994 (0.987-1.002)	--	--
SA	--	1.020 (1.006-1.034)	1.020 (1.006-1.034)	--	1.004 (0.996-1.012)	1.004 (0.996-1.012)
IDW	--	1.017 (1.004-1.030)	1.018 (1.004-1.032)	--	0.995 (0.987-1.003)	0.995 (0.987-1.003)
KRIG	--	1.018 (1.004-1.032)	1.017 (1.003-1.031)	--	0.994 (0.986-1.003)	0.997 (0.989-1.005)
CMAQ	--	1.009 (1.003-1.015)	1.011 (1.004-1.018)	--	1.002 (0.998-1.005)	1.003 (1.000-1.007)
CMAQKR	--	1.017 (1.004-1.030)	1.010 (0.995-1.024)	--	0.994 (0.987-1.002)	0.995 (0.988-1.003)
CMAQDF	--	1.019 (1.007-1.030)	1.009 (0.994-1.023)	--	1.000 (0.993-1.007)	0.997 (0.989-1.005)
RR (95% CI) for OC per 1 µg/m³ increase						
CM	1.008 (1.004-1.013)	--	--	1.000 (0.998-1.003)	--	--
SA	--	1.010 (1.004-1.016)	1.010 (1.004-1.016)	--	1.002 (0.999-1.005)	1.002 (0.999-1.005)
IDW	--	1.008 (1.004-1.013)	1.010 (1.005-1.016)	--	1.000 (0.998-1.003)	0.998 (0.995-1.001)
KRIG	--	1.009 (1.004-1.013)	1.009 (1.004-1.015)	--	1.000 (0.998-1.003)	0.997 (0.994-1.000)
CMAQ	--	1.005 (1.002-1.007)	1.005 (1.002-1.007)	--	1.001 (1.000-1.003)	1.001 (1.000-1.003)
CMAQKR	--	1.008 (1.004-1.013)	1.007 (1.002-1.012)	--	1.000 (0.998-1.003)	1.000 (0.998-1.003)
CMAQDF	--	1.010 (1.003-1.017)	1.010 (1.003-1.017)	--	1.003 (0.999-1.007)	0.997 (0.992-1.001)

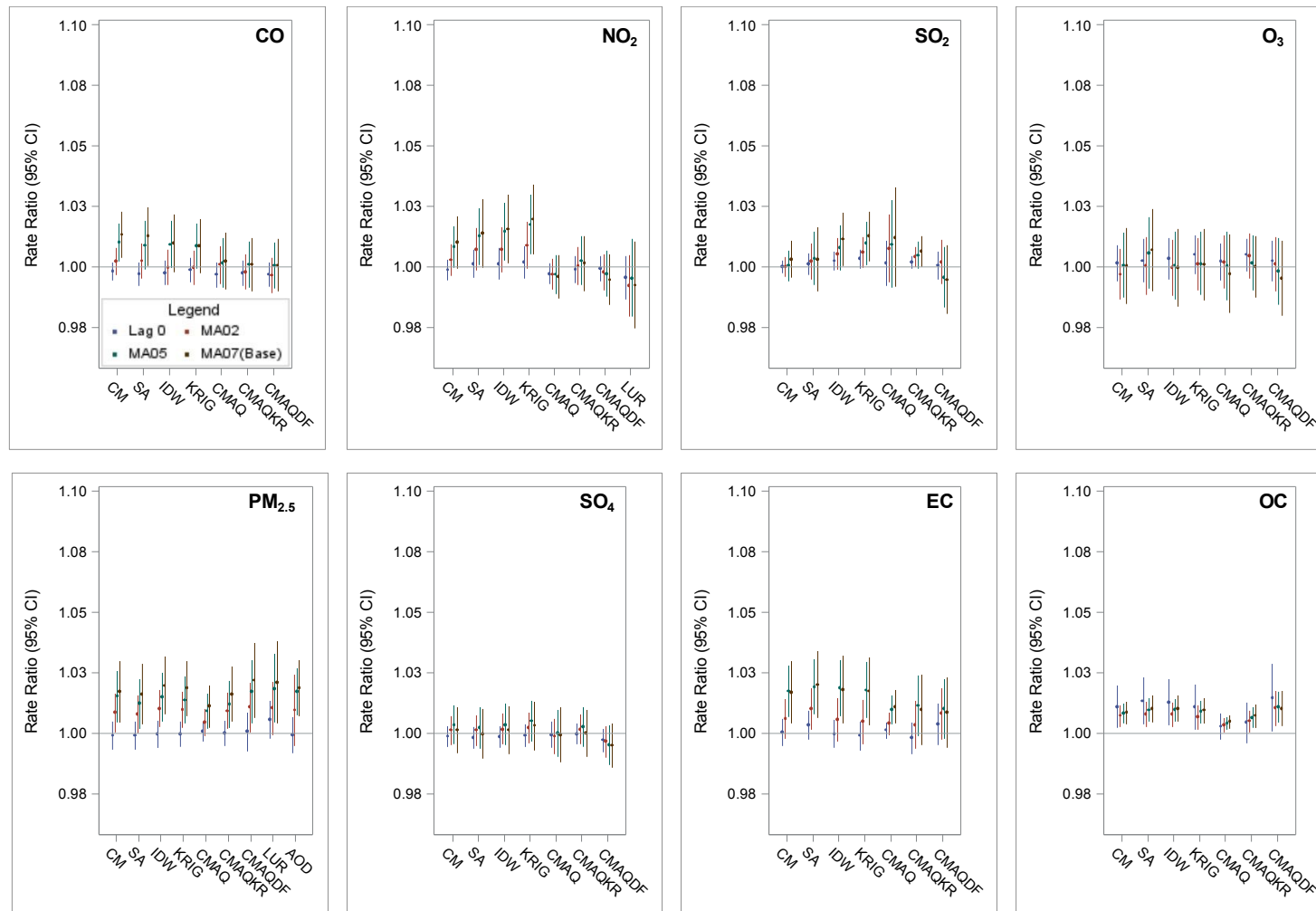
¹Exposure metrics included: Central Monitor (CM), Site Average (SA), Inverse Distance Weighting (IDW), Kriging (KRIG), Community Multiscale Air Quality model (CMAQ), CMAQ-kriging adjustment with observations (CMAQKR), CMAQ combined with monitoring data in a data fusion (CMAQDF), Land Use Regression (LUR), and satellite Aerosol Optical Depth (AOD).

²Study approaches included assigning: (1) Central Monitor values to all 192 ZIP codes by day, (2) Central ZIP Code values for non-CM exposure metrics to all 192 ZIP codes by day, and (3) the Full Spatial information of the exposure metrics to the 192 ZIP codes by day.

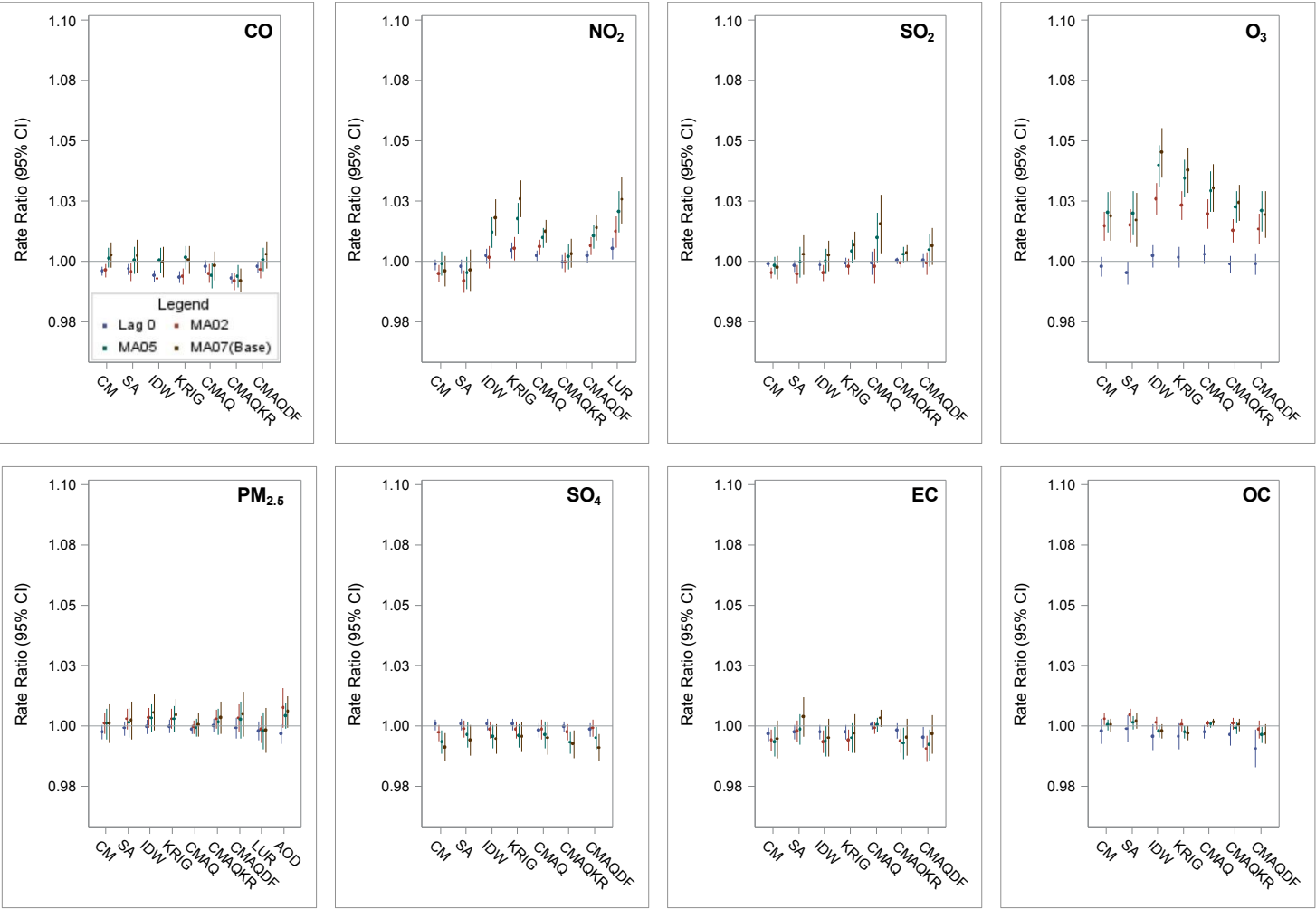
³All models controlled for time and meteorology as specified by the base model.



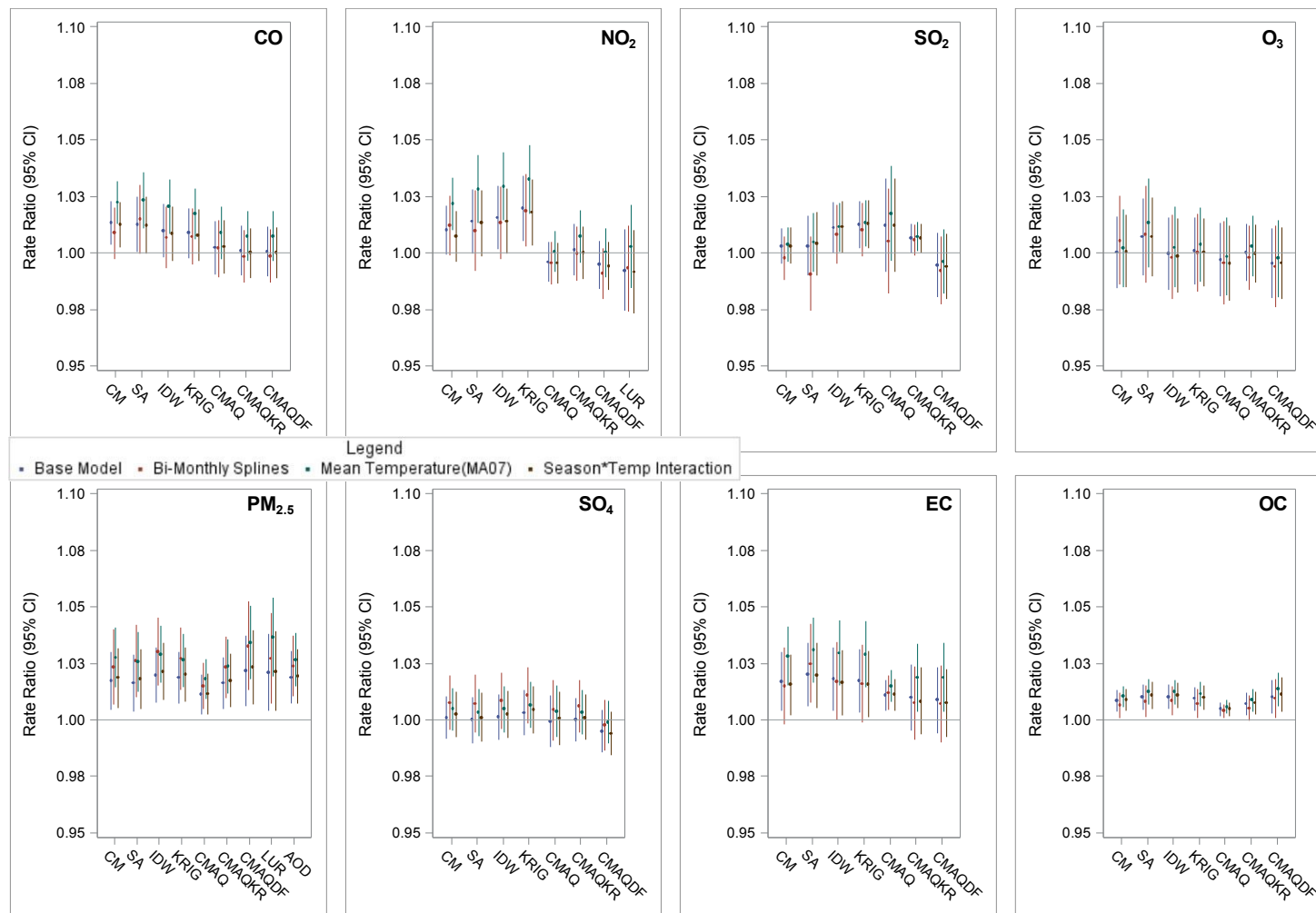
Supplemental Figure S1. Distribution of a) ED visit counts across 20-county Atlanta ZIP codes for cardiovascular diseases (left) and respiratory diseases (middle), and b) ZIP code-specific Pearson correlations of ED visit counts between the two outcomes (right).



Supplemental Figure S2. Associations of air pollution and ED visits for cardiovascular diseases, comparing among exposure windows and exposure metrics at full spatial resolution. Exposure metrics included: Central Monitor (CM), Site Average (SA), Inverse Distance Weighting (IDW), Kriging (KRIG), Community Multiscale Air Quality model (CMAQ), CMAQ-kriging adjustment with observations (CMAQKR), CMAQ combined with monitoring data in a data fusion (CMAQDF), Land Use Regression (LUR), and satellite Aerosol Optical Depth (AOD). Study approach utilized the Full Spatial information of data at 192 ZIP codes. Exposure windows included lag 0, the moving average (MA) of lag days 0-2, 0-5, and 0-7. All models controlled for time and meteorology akin to the base model. Rate ratios (RR) and confidence intervals (CI) scaled to standard units: per 250 ppb CO, 10 ppb NO₂, 4 ppb SO₂, 15 ppb O₃, 5 µg/m³ PM_{2.5}, 1 µg/m³ SO₄²⁻, 0.5 µg/m³ EC, and 1 µg/m³ OC.

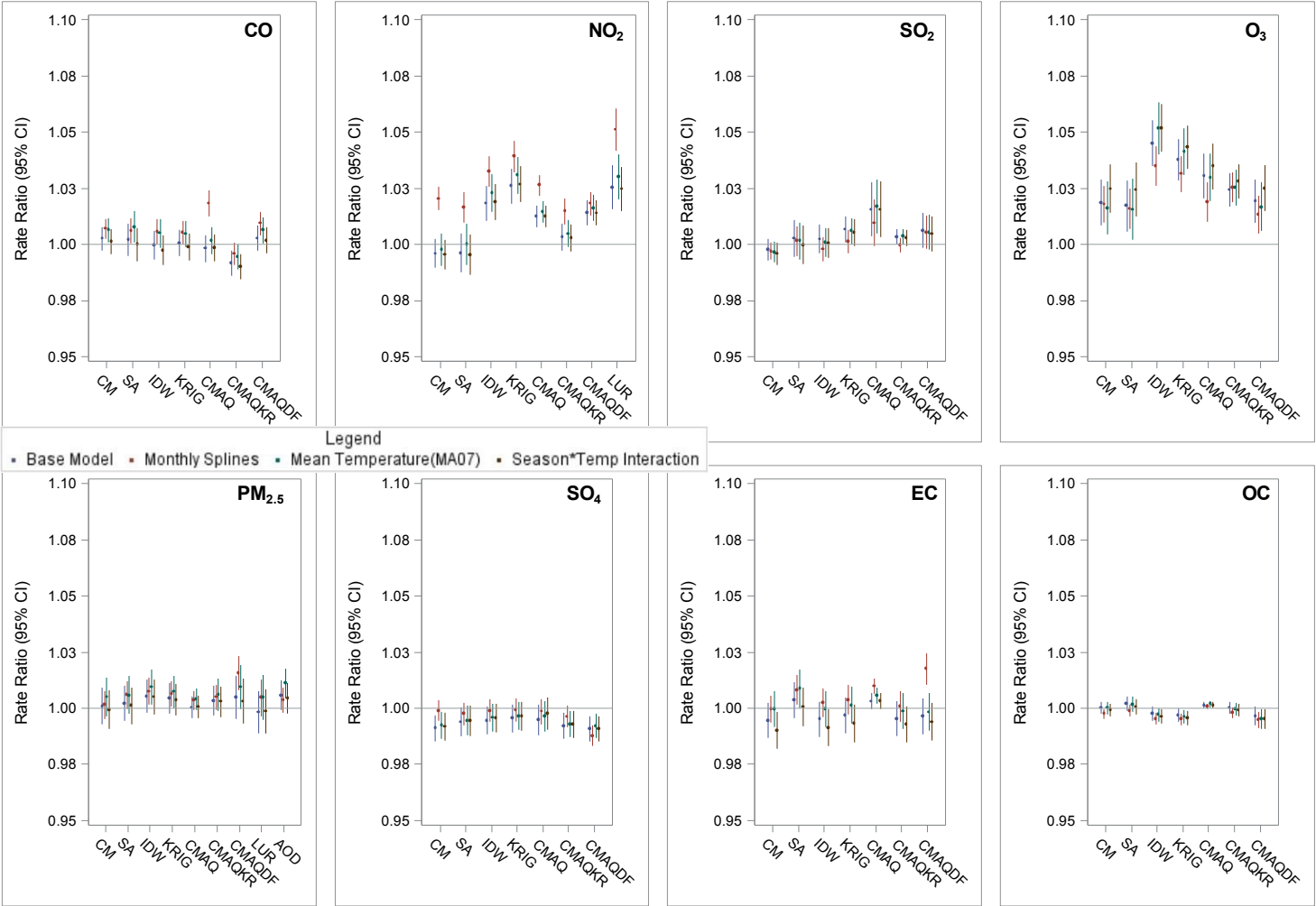


Supplemental Figure S3. Associations of air pollution and ED visits for respiratory diseases, comparing among exposure windows and exposure metrics at full spatial resolution. Exposure metrics included: Central Monitor (CM), Site Average (SA), Inverse Distance Weighting (IDW), Kriging (KRIG), Community Multiscale Air Quality model (CMAQ), CMAQ-kriging adjustment with observations (CMAQKR), CMAQ combined with monitoring data in a data fusion (CMAQDF), Land Use Regression (LUR), and satellite Aerosol Optical Depth (AOD). Study approach utilized the Full Spatial information of data at 192 ZIP codes. Exposure windows included lag 0, the moving average (MA) of lag days 0-2, 0-5, and 0-7. All models controlled for time and meteorology akin to the base model. Rate ratios (RR) and confidence intervals (CI) scaled to standard units: per 250 ppb CO, 10 ppb NO₂, 4 ppb SO₂, 15 ppb O₃, 5 µg/m³ PM_{2.5}, 1 µg/m³ SO₄²⁻, 0.5 µg/m³ EC, and 1 µg/m³ OC.



Supplemental Figure S4. Associations of air pollution and ED visits for cardiovascular diseases, comparing among different temporal controls specification and exposure metrics at full spatial resolution. Exposure metrics included: Central Monitor (CM), Site Average (SA), Inverse Distance Weighting (IDW), Kriging (KRIG), Community Multiscale Air Quality model (CMAQ), CMAQ-kriging adjustment with observations (CMAQKR), CMAQ combined with monitoring data in a data fusion (CMAQDF), Land Use Regression (LUR), and satellite Aerosol Optical Depth (AOD). Study approach utilized the Full Spatial information of data at 192 ZIP codes. Models tested different specifications of temporal controls compared to the base model: controlling for time using parametric cubic splines with two knots per month; controlling for temperature with cubic terms for the moving average of mean temperature (lag days 0-7);

and adding a product term for season*temperature. Rate ratios (RR) and confidence intervals (CI) scaled to standard units: per 250 ppb CO, 10 ppb NO₂, 4 ppb SO₂, 15 ppb O₃, 5 µg/m³ PM_{2.5}, 1 µg/m³ SO₄²⁻, 0.5 µg/m³ EC, and 1 µg/m³ OC.



Supplemental Figure S5. Associations of air pollution and ED visits for respiratory diseases, comparing among different temporal controls specification and exposure metrics at full spatial resolution. Exposure metrics included: Central Monitor (CM), Site Average (SA), Inverse Distance Weighting (IDW), Kriging (KRIG), Community Multiscale Air Quality model (CMAQ), CMAQ-kriging adjustment with observations (CMAQKR), CMAQ combined

with monitoring data in a data fusion (CMAQDF), Land Use Regression (LUR), and satellite Aerosol Optical Depth (AOD). Study approach utilized the Full Spatial information of data at 192 ZIP codes. Models tested different specifications of temporal controls compared to the base model: controlling for time using parametric cubic splines with two knots per month; controlling for temperature with cubic terms for the moving average of mean temperature (lag days 0-7); and adding a product term for season*temperature. Rate ratios (RR) and confidence intervals (CI) scaled to standard units: per 250 ppb CO, 10 ppb NO₂, 4 ppb SO₂, 15 ppb O₃, 5 µg/m³ PM_{2.5}, 1 µg/m³ SO₄²⁻, 0.5 µg/m³ EC, and 1 µg/m³ OC.